

MG-1S农业植保机定损

MG-1S Damage Assessment



外观检测

Visual Inspection

功能检测

Functional Inspection



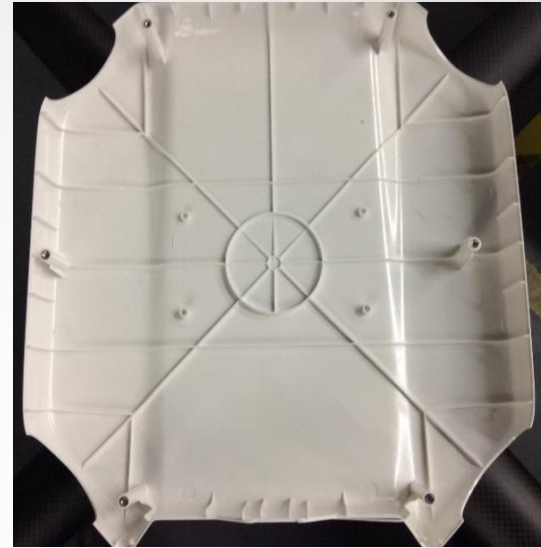
外观检查

Visual Inspection



1. 使用2.5内六角螺丝刀卸下所有桨叶，并检查桨叶外观是否破损、断裂；
2. 转动电机，检查电机外观是变形、刮伤；并检查电机是否堵转、卡转；
3. 垂直拉动电机转子，检查电机旷量是否正常；

1. Rotate the motors and check whether they are misshapen, there are scratches on them and they are obstructed.
2. Remove all propellers with a 2.5 hexagonal socket screwdriver and check whether the propellers are damaged or cracked.
3. Pull the rotors vertically and check whether the gap between the stator and rotor is appropriate.

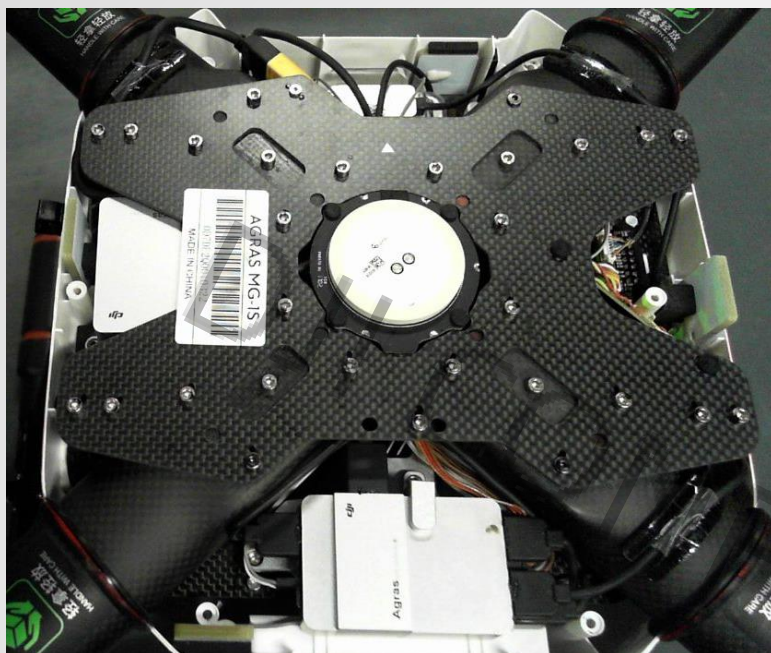


- 4.检查力臂是否断裂，转动力臂，检查力臂是否变形；
- 5.取下固定上壳的螺丝，将上壳取下；
- 6.检查上下壳是否变形破损，检查机身内部螺丝柱是否损坏；

4.Check whether the arms crack. Rotate the arms and check whether they are misshapen.

5.Loosen the screws for fixing the upper shell and remove the upper shell.

6.Check whether the upper shell and lower shell are misshapen or damaged and check whether the screw bases in the airframe are damaged.

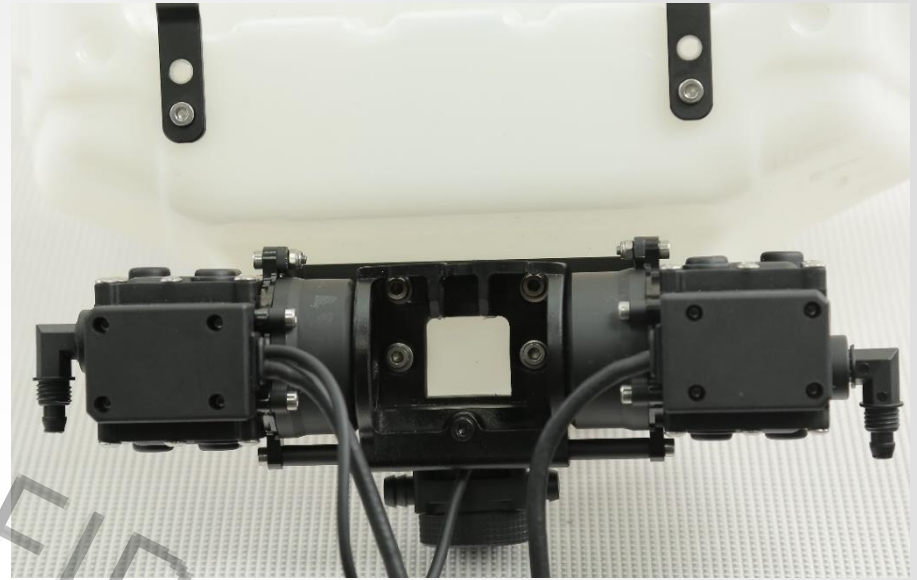


- 7.检查椭圆管臂是否变形断裂，检查折叠位置和密封圈是否变形损坏；
- 8.检查主控、PMU、GPS等功能模块外观是否损坏；
- 9.检查GPS背面的防水贴是否变红；

7.Check whether the oval-shaped arms are misshapen or crack and whether the folding positions and sealing rings are misshapen or damaged.

8.Check whether the functional modules such as the main controller, PMU and GPS are damaged.

9.Check whether the water damage sticker on the back of the GPS module is red.



10.检查左右脚架上的碳管是否断裂、损坏；

11.检查水箱是否破损、外观是否有裂纹；

12.检查水泵外观是否变形损坏；

10. Check whether the carbon tubes of the left and right landing gears crack or are damaged.

11. Check whether the Spray Tank is damaged and there are cracks on it.

12. Check whether the Spray Tank is misshapen or damaged.



13.检查雷达外观是否变形损坏；

14.检查水管、喷头是否断裂破损；

13.Check whether the radar is misshapen or damaged.

14.Check whether the hoses and sprinklers crack or are damaged.

功能检查

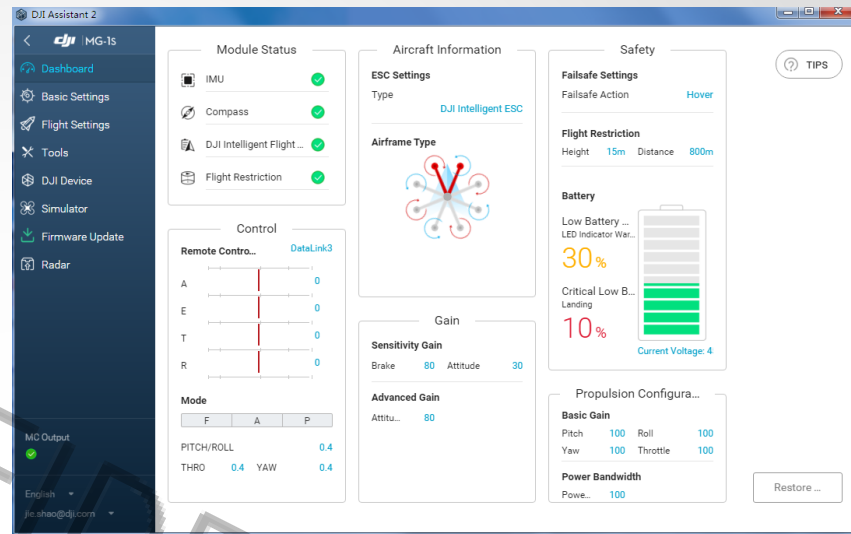
Functional Inspection



1. 将飞行器放在水平面上，插上开机检测尾灯LED是否闪烁；
2. 连接农业机下壳的导数据USB接口，打开DJI Assistant软件，点击MG-1S进入调参内部界面；

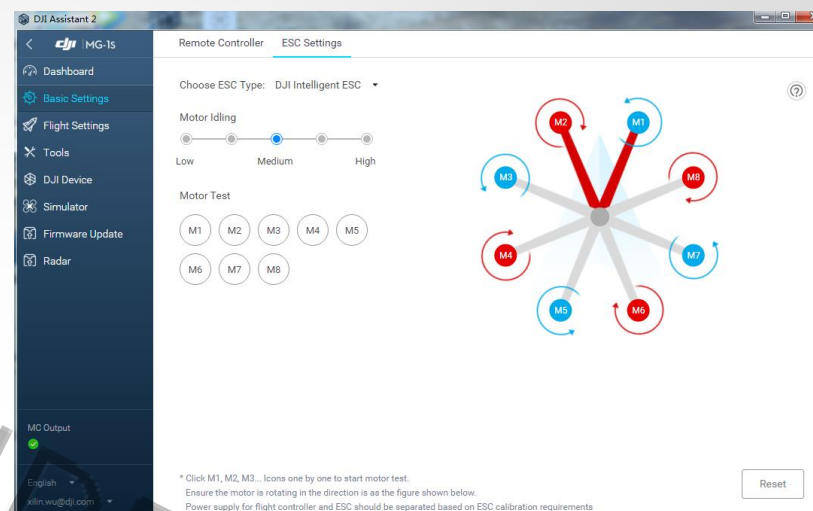
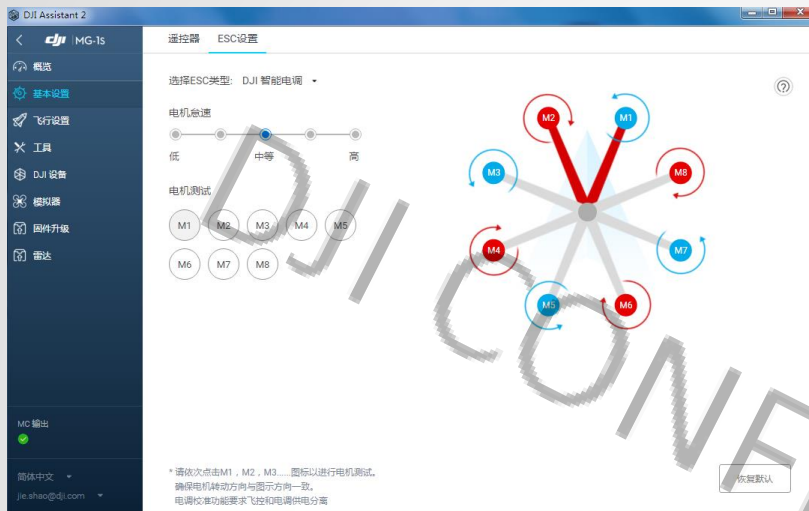
1. Place the MG-1S on a level surface, power on the MG-1S and check whether the rear LED is blinking.

2. Insert the USB cable to the USB port on the lower shell of the MG-1S, launch DJI Assistant 2 and click MG-1S.



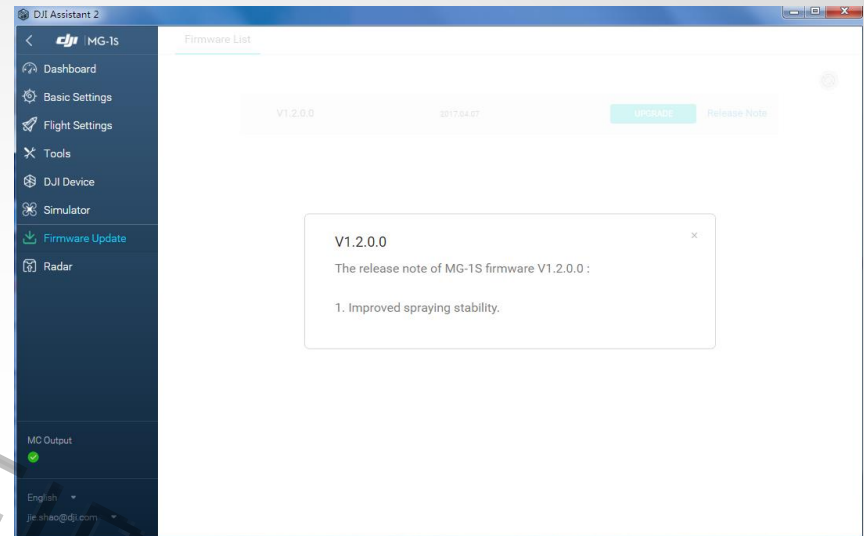
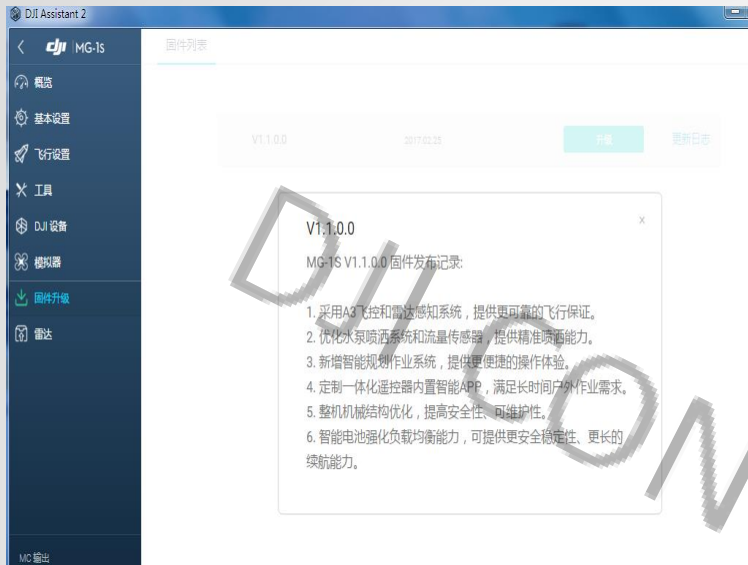
3.概览->模块状态可以检查到飞行器IMU、指南针、电池等功能模块是否正常

3.Click “Dashboard” and check whether the functional modules such as IMU, compass and battery are functioning normally under Module Status.



4.基本设置 ->ESC设置依次检查M1-M8电机的转向，其中M1/3/5/7电机为逆时针转，M2/4/6/8号电机为顺时针转；

Click “Basic Settings” and “ESC Settings” and check the rotation direction of the motors 1 to 8 one by one. Motor 1, motor 3, motor 5 and motor 7 are rotating counterclockwise while motor 2, motor 4, motor 6 and motor 8 are rotating clockwise.



5.固件升级 -> 固件列表中检查飞行器是否需要升级，可在更新日志内部检查新固件所更新的内容；

5.Click “Firmware Update” and “Firmware List” and check whether the MG-1S needs to be updated. Click “Release Note” to check what’s new of the new firmware.



6. 打开遥控器，检测飞行器能否和遥控器正常连接；

Lightbridge 2 对频：

(1) 开机状态下，直接按下接收机的对频键，接收机指示灯为红灯闪烁，对频成功后为绿灯常亮；(需要拆开外壳)

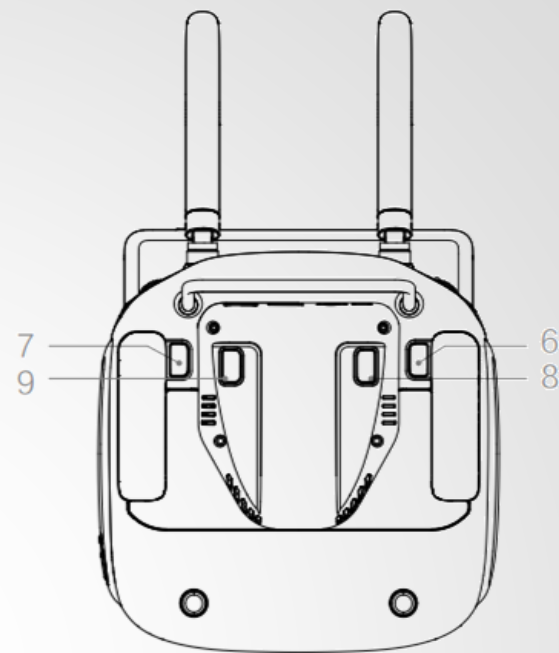
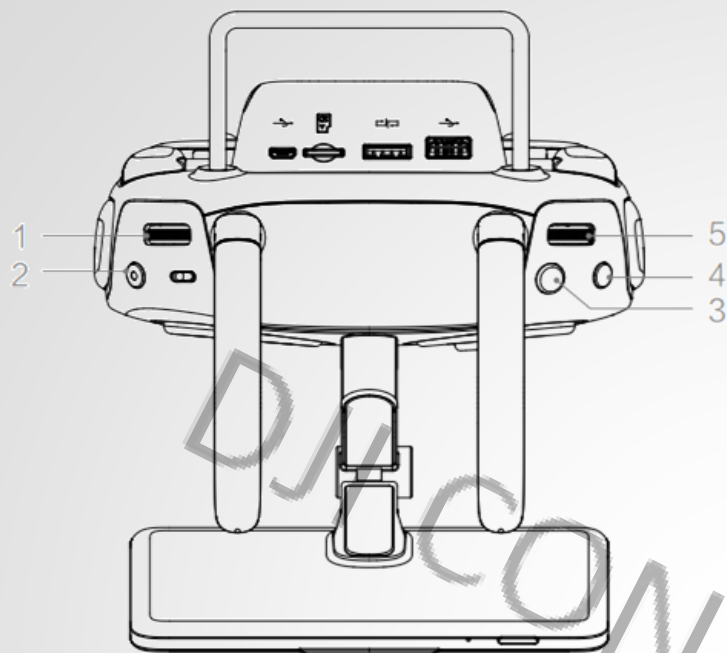
(2) 开机状态下，直接按下中心板下方的对频按键，飞行器就进入对频状态

4. Turn on the remote controller and check whether the MG-1S is connected correctly with the remote controller.

How to Link the Lightbridge 2 with the MG-1S:

(1) With the Lightbridge 2 powered on, press the Link Button on the receiver and the LED on the receiver will blink red. After linking is successful, the LED on the receiver will be solid green. (The shell needs to be removed.)

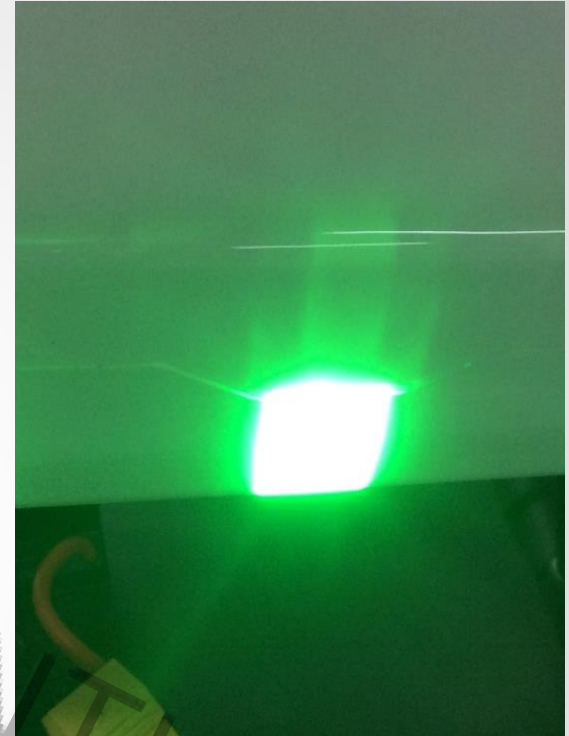
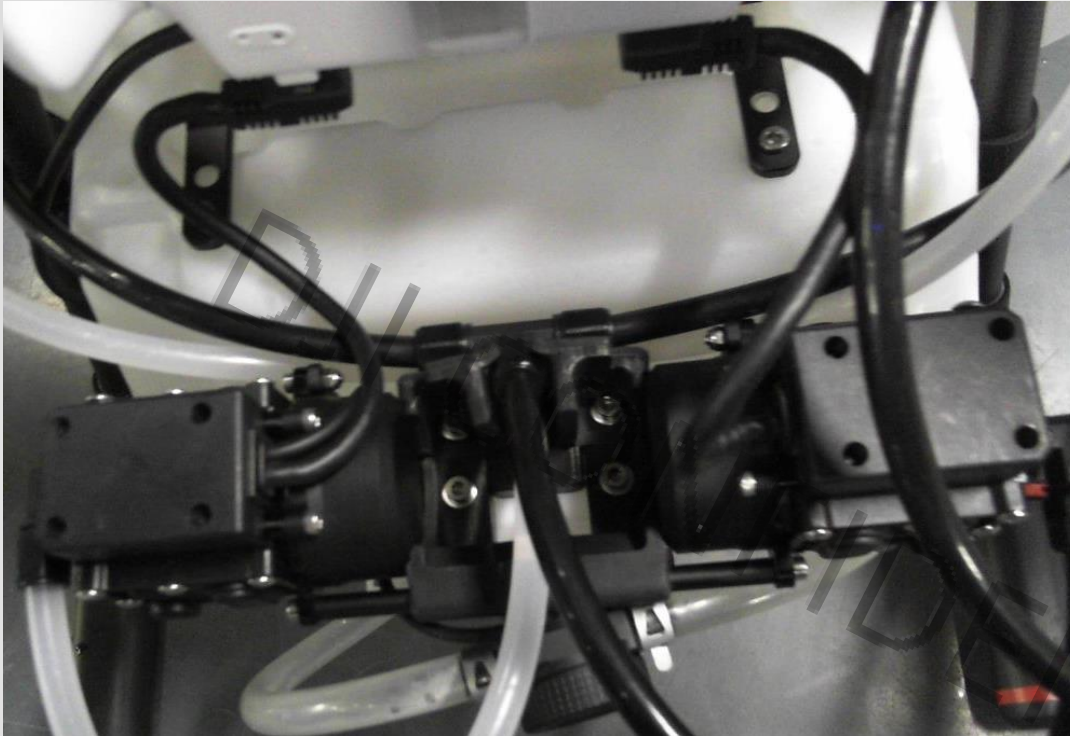
(2) With the MG-1S powered on, press the Link Button under the central board and the MG-1S will link with the Lightbridge 2.



遥控器对频：开机状态下，同时按住C1+C2+喷洒键（图中6+7+2）听到遥控器响声后，按下飞行速度设置转盘（5），此时遥控器指示灯亮蓝灯常亮，进入对频状态，对频成功后指示灯为绿灯常亮

How to Link the Remote Controller with the MG-1S

With the remote controller powered on, press the C1 Button, C2 Button and Spray Button (marked as 6, 7 and 2 in the figure). After a sound is emitted by the remote controller, press the Flying Speed Dial (marked as 5 in the figure) and the Status LED will be solid blue, indicating that it is linking with the MG-1S. After linking is successful, the Status LED will be solid green.



7. 检查水泵接口是否损坏变形；启动水泵，检查水泵电机是否启动；（水泵）
8. 断开电池，检查尾灯绿灯是否绿灯常亮。（断电续航）

7. Check whether the ports of the delivery pump are misshapen or damaged. Turn on the delivery pump and check whether its motor can start normally. (This is to check whether the delivery pump works normally.)
8. Remove the batteries and check whether the rear LED is solid green. (This is to check whether the reserved battery module works normally.)

9. 动力饱和要留意以下几点：

- ① 飞行器在高海拔地区飞行，飞手打杆猛烈；
- ② 在飞行器上桨叶是否有粘附物，是否破损等；
- ③ 飞行过程中遇到强风，使飞行器姿态不稳；
- ④ 电机电调是否故障，留意飘窗信息。

（PS：如果有风，姿态不稳，又满载，飞行速度要设小点，动力饱和的提示才会消失，也相对安全。）

9. When the DJI MG app displays a warning stating that motors are overloaded, check the following:

- ① Whether the MG-1S is flying in a high-altitude region and the control sticks are pushed drastically.
- ② Whether there are unwanted items on the propellers, or the propellers are damaged, etc.
- ③ Whether the aircraft's attitude is unstable during flight because the winds are strong.
- ④ Whether the ESCs are malfunctioning. Pay attention to the warnings displayed in the app.

(Note: If the winds are strong, the aircraft's attitude is unstable, and motors are overloaded, slow down the flight speed and the warning will disappear. This also guarantees flight safety.)

10. 水泵不工作喷头不全部喷洒需注意以下几点：

- ① 水泵喷洒系统重新校准（药箱加满10L的药，上天用智能作业或航线规划作业进行喷洒系统校准）；
- ② 拆开流量计，检查清洗小叶轮，重新安装；
- ③ 检查药箱滤网是否堵塞，喷头排气阀的橡胶垫是否由于使用乳油药品导致膨胀变形（变形的话可用清水泡洗然后晾干）。
- ④ 检查水泵排线是否没有脱落，或水泵压力计故障导致水泵异常，如果是可以更换水泵进行测试；

10. When the delivery pump does not work and no sprinkler sprays pesticides, perform the following:

- ① e-calibrate the spraying system (fill the spray tank with 10 L pesticides, take off the aircraft, enable Smart Operation Mode or Field Plan Mode, and calibrate the spraying system).
- ② Disassemble the flowmeter, check and clean the impeller, and then reassemble the flowmeter.
- ③ Check whether the filter mesh in the spray tank is obstructed, and whether the rubber pad in the outlet valve of the sprinkler is swollen and misshapen due to emulsifiable concentrate formulations (if the rubber pad is misshapen, immerse it in clean water for a while, take it out and dry it.)
- ④ Check whether the flat cable of the delivery pump comes off, or check whether the delivery pump does not work because the manometer of the delivery pump is malfunctioning. If available, replace the delivery pump with another normally functioning delivery pump to check whether the original delivery pump is malfunctioning.

11. MG-1S雷达链路异常解决方法：

- ① 检查线路是否出现腐蚀、破损；
- ② 检查线路连接处是否进水，短路；
- ③ 如若出现因腐蚀造成的破损请记录使用的药品名称；
- ④ 每日作业结束后请，及时对雷达及线路进行清洗，包括雷达背面；
- ⑤ 断电重新插入雷达连接线，开机重启，检查是否线路松动；
- ⑥ 把原先在下边的定高雷达用其他雷达（前后雷达）替换，把线端口用酒精清洗再插上，拆下的定高雷达清洗干净替换到前后雷达位置上。（PS：调换位置后雷达需要重新定义，具体方法是，轻按指示灯边上按钮，等指示灯跳到对应的颜色（前蓝中黄后绿）后松开，重新上电，看看是不是单闪刚刚设置的颜色。）

11. If there is an error with the MG-1S radar link, perform the following:

- ① Check whether the cables are corroded or damaged.
- ② Check whether the cables connections come into contact with water, or are short-circuited.
- ③ If the damage is caused by corrosion, record the name of the pesticides.
- ④ Clean the radar and cables in a timely manner, including the back of the radar, each day after the operation is finished.
- ⑤ Power off the aircraft, remove and insert the radar connection cables again, power on the aircraft, and check whether the cables are loose.
- ⑥ Replace the lower radar (for altitude stabilization) with the front radar or rear radar, clean the connectors of the cables with alcohol, connect the cables, clean the lower radar and mount it to the front radar or rear radar's position. (Note: The radar needs to be redefined after being replaced. Press and hold the button besides the LED, release it when the LED blinks a dedicated color (blue for front radar, yellow for lower radar, and green for rear radar), power on the radar, and check whether the LED blinks the dedicated color.)

12. MG-1S中心板和电池连接失败，目前已知四大类原因，请排查：

- ① 电池数据线接触不良，请更换电池排除电池因素。
- ② 检查接线插口松动，请检查线路和插头。
- ③ 雷达进水短路，请拔下雷达线，烘干雷达。
- ④ 中心板硬件损坏，需更换中心板。

12. There are four causes for the issue where the MG-1S central board fails to connect to the battery. Check the following:

- ① The battery's USB cable is in poor contact. Replace the battery with another normally functioning battery to check whether the original battery is malfunctioning.
- ② The cables connections are loose. Check the cables and connectors.
- ③ The radar is water damaged and short-circuited. Remove the radar cable and dry out the radar.
- ④ The central board is damaged. Replace the central board.

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